Stochastic Simulation Assignment 3
CT scan simulation

L. Troiakova (3377822) & W. Hui (3307530)

University of Twente

June 22, 2026

1 Introduction

In this report we will present a solution to multiple problems for a radiology department. There they currently have 2 CT scanners, for which we are asked to simulate the current situation. The radiology department is using a specific policy for patient scheduling, that we will describe closer in the section Problem description in words. Besides the patients that need to be scheduled, the outpatients, there is two more types of patients, emergency and inpatient, which both arrive randomly, but with different rates. Emergency patients also have a priority in scanning. There is a waiting room in the department that currently has a capacity to accommodate 3 people waiting. If there is more patients waiting, they must wait outside of the room.
There is multiple questions we want to answer throughout this report. Firstly, we would like to make an analytical capacity check. This check will answer the question whether the two CT scanners are sufficient to treat all the patients. Secondly, we will examine the system using discrete event simulation. This simulation will give us answers to questions like, whether the capacity of the CT scanners is used to its total potential. We also calculate what is the time between an outpatient calling and their scheduled appointment, what is the waiting time for both emergency patients and outpatients after their arrival, and the fraction of patients that have to wait outside of the waiting room. Lastly, we calculate what is the percentage of the inpatients that cannot be scanned before 16:00 on the same working day they called for their appointment.
In this report we at first describe the problem closely, then introduce all the variables that will be used, as well as the mathematical formulas how we can derive these variables. In a separate section we describe the simulation model and present the pseudo-code that describes it. We perform an analysis on the capacity. After this, we present the results that follow from both the simulation and analysis.

2 Problem Description

In the radiology department there are two CT scanners that will be open from 8:00 to 16:00 each day. This time we also call office or working hours in below parts. Outside of the office hours, one of the CT scanners remains available. For clarity we assigned index 1 and 2 to the two scanners. The scanner 1 works all the time inside and outside office hours during evenings, nights, and weekends, and the scanner 2 works only in office hours.
There is three types of patients arriving into the CT department. The first type is emergency patients. These patients have a priority of admission over other types of patients. The second type of patients are outpatients. These call in advance and schedule an appointment. Currently, the radiology department has a particular policy for scheduling outpatients. The last type of patients are inpatients, which arrive randomly. Outpatients and inpatients are treated in order of arrival.
In the moment the radiology department is not using any dynamic blueprint. In this report we will simulate the current situation, which will give us a mean to analyze it and give numerical results to multiple problems. We want to observe to what extend are the scanners used throughout a day, what are the average waiting times for different types of patients, whether the design of the waiting room, which right now has 3 waiting spaces is sufficient, and also analyze whether using 2 scanners during the working hours and 1 scanner outside of working hours is enough.

2.1 Variables

Since the duration of scanning patients are in the uniform distribution U(10,19).

$$E[t] = \frac{10+19}{2} = 14.5min = 0.2417h \tag{1}$$

$$\mu = \frac{1}{0.2417} = 4.138 \tag{2}$$

The emergency arrival rate

$$\lambda_E = \frac{24}{24} = 1 \tag{3}$$

The inpatient arrival rate is depending on the hour in the day. When it is outside the office hours on a weekday is all constant.

$$\lambda_I = \frac{6}{16} = 0.375 \tag{4}$$

In the working hours it follows sine.

$$A = \frac{2\lambda_I + 6(\lambda_I + A\frac{2}{\pi})}{8} \tag{5}$$

So the when it is outside working hours the rate λ_I remains constant.

$$\lambda_I(t) = 0.375 \tag{6}$$

And during office hours we have that going to peak twice around 10:30 AM and around 13:30 PM with time changing to λ_I.

$$\lambda_I(t) = 0.375 + 1.5\pi sin(\frac{\pi(t-9)}{3}) \tag{7}$$

$$\lambda_I(t) = 0.375 + 1.5\pi sin(\frac{\pi(t-12)}{3}) \tag{8}$$

as peak rates.

The outpatient call rate

$$\lambda_{OC} = \frac{23}{8} = 2.875 \tag{9}$$

variables	value	unit
μ	4.138	service rate
λ_E	1	per hour
$\lambda_{I_{day}}$	2.625	average per hour
$\lambda_{I_{night}}$	0.375	average per hour
λ_O	2.875	per hour
λ_P	8.3	per hour

TABLE 1: Values

3 Simulation Model

In the following we write the emergency type in short to be E, inpatient to be I, and outpatient to be O. The simulation runs for 200 weeks for precision check.

3.1 Assumptions

We model this system as a discrete simulation, the time is measured in continuous hours. We assume the simulation start at Monday 00:00 and assign this day as day 0 the initial day. The office hours are [8, 16] in hours from Monday to Friday.
We assume the emergency and inpatient will also arrive outside the office hours which are independent Poisson processes.The arrivals for emergency patient and inpatient are independent Poisson processes, which means the arrival is exponentially distributed. The arrivals for inpatients are randomly drawn at a rate dependent on time. The probability of arriving in office hours follows a average fraction that come in office hours : non office hours = 21:6
Inpatients arrive immediately. The arrivals for outpatients are not Poisson process but being scheduled. The arrival of the outpatients appointment calls are exponential distributed. They will show up with probability 0.84.
Outpatient slots are managed on a weekly cycle. If no slot is available in the current week before Friday 16:00, the patient is placed on a waiting list and schedule to the following week.
The scanning happen before 16:00 and continue after 16:00 will not be closed and the second scanner will be closed after finish this scanning.
The duration of CT scanning is uniformly distributed on [10,19] in minutes for all patients
We assume the time slots within one hour are equally divided this hour, not all scheduled period are 15 minutes.
The limitation of chairs does not mean the limitation of the queue, which means we allow patient more than 3 in the queue.

3.2 Events

The simulation mainly consist of two types of events, arrivals and executions.

3.2.1 Arrivals

There are three types of patients with different arriving rules.All three arrivals assign patients and either route them to a scanner immediately or add them to a queue.

Algorithm 1 Arrival

Emergency arrival:
schedule next emergency arrival at time (t + Exp(λ_E))
assign the patient with type emergency ▷ Use Patient above
record arrival time ▷ check if is in office hours(time and day)
if find free scanners **then**:
    record the status ▷ waiting time = 0
    start scanning this patient at current scanner
else:
    add this patient to emergency queue ▷ priority queue
end if
Inpatient arrival:
schedule next inpatient arrival at time (t + Exp(λ_I)) ▷ λ_I is changing through time with sine function
assign the patient with type inpatient
record arrival time
if find free scanners **then**:
    record the status
    start scanning this patient at current scanner
    call the next patient in the queue immediately
else:
    add this patient to normal waiting queue ▷ check if the queue length $>$ 3 (for chair limitation)
end if
Outpatient arrival:
schedule next call of outpatients at (t + Exp(λ_O)) ▷ in the office hours
assign the patient with type outpatient
find the earliest free slot from the next day to the end of the week ▷ If later than Friday then schedule for next week
if the outpatient show up **then**: ▷ with probability p=0.84
    record arrival time as scheduled time ▷ assume if they come then they are on time
    if find free scanners **then**:
        record the status
        start scanning this patient at current scanner
    else:
        add this patient to normal waiting queue ▷ check if the queue length $>$ 3 (for chair limitation)
    end if
end if

3.2.2 Scannings

The scanning process handles assigning patients to scanners, running the scan, and freeing the scanner when complete the service.

Algorithm 2 Scanning

draw random scanning duration time
set the current status of the scanner to be busy
record current time t ▷ check if the starting time and ending time to be in the office hours
schedule ending time (t + duration time)
if is using scanner 2 and exceed office hours **then**
    change the closing time of scanner to be the ending time of the scan
end if ▷ do not interrupt and force to close
if emergency queue in not empty **then**: ▷ pick next patient to start next scanning
    scan the emergency patient with index 0
else:
    scan the patient in normal queue with index 0
end if

3.3 Verification

Since it is almost impossible to calculate an accurate value for the whole model by hand and we only aim to check the simulation is working correctly. We verify the model by checking some values are relating correctly and run a simple state that can be calculated manually to check if it is valid.
We test the validation of the model by changing the input values to extreme values. If we set the number of chairs in the waiting room to be 0, the number of patients outside waiting room is equal to the number of total patients in the queue. And the total generated arrivals equal to the total completed service in the system at the end of simulation. If we set arrival rates $\lambda = 0$, the utilization of the scanner will also be 0.
Then we know the relations between given values are valid. We now examine the model with an example that we could easily check the value. In this case we structure a M/M/2 sample with only emergency patients($\lambda_E = 1$) by setting $\lambda_I = 0$, $\lambda_O = 0$. The number of scanners c=2. The theoretical mean queue length and mean waiting time in the queue will be calculated by Erlang formula.

$$L_q = \frac{P_Q \cdot \rho}{1-\rho} = \frac{\lambda^3 \mu}{(2\mu - \lambda)^2}$$

where $\rho = \frac{\lambda}{c\mu} < 1$and $P_Q = \sum_{n=2}^{\infty} P_n = \frac{1}{c!}\frac{\lambda}{\mu}\frac{1}{1-\rho}P_0$ and queue length equals $\sum_{n=c}^{\infty}(n-c)P_c$.

$$L = L_q + \frac{\lambda}{\mu}$$

$$W_q = \frac{L_q}{\lambda}$$

and according to the Little's Law $L = \lambda W$, the total time of a scan from arrival to complete should be

$$W = \frac{L}{\lambda} = W_q + \frac{1}{\mu}$$

with the parameter $\mu = 4.138$, calculated the theoretical values $\rho = 0.121$, $P_0 = 0.785$, $P_Q = 0.0264$ and $W_q = 0.00373$, which meets the simulated values with confidence interval under the same simplified condition. so it proves the event loops and schedule logic are valid, and suit for multiple scanners servers. The model is valid.

4 Analysis

4.1 Analytical capacity check

Since the ct simulation is a simulation with multiple independent queues and shared servers. We want that the λ to be # patients per hour. so the capacity c is

$$c = \frac{\lambda_{total}}{(\#servers) \cdot \mu} \tag{10}$$

During the working time we have, where

$$\lambda_E = 1$$

$$\lambda_{IP} = 21/8 = 2.625$$

$$Max(\lambda_{OP}) = \frac{Max(\#appointements)}{\#hours} = \frac{28}{8} = 3.5$$

$$\lambda_{total} = \lambda_E + \lambda_{IP} + \lambda_{OP} = 7.125$$

and out of the working time we have,

$$\lambda_E = 1$$

$$\lambda_{IP} = 6/16 = 0.375$$

$$\lambda_{total} = \lambda_E + \lambda_{IP} = 1.375$$

In office hours day time there will be two scanners working and in the non office hours night time there is only canner 1 working. so the intensity per scanner will be

$$c_{day} = \frac{\lambda_{total}}{(\#servers) \cdot \mu} = \frac{7.125}{2 \cdot 4.138} = 0.861$$

$$c_{night} = \frac{\lambda_{total}}{(\#servers) \cdot \mu} = \frac{1.375}{1 \cdot 4.138} = 0.332$$

and we compare these to 1. Since both $c < 1$ then we have that stable. and the capacity of scanners in working time is $c_{day} = 0.861$,and is $c_{night} = 0.332$ in non working hours.

4.2 Data that you collect during the simulation

For the scanner utilization, it is a time average value so we collected the time duration of scanners working and the actual absolute clock time the scanners start and end working.
The other metrics we need to collect is recorded dependent on patient. We collect the following values.

data	used to calculate
working time for scanner 1 during office hour	office hour utilization of scanner 1
working time for scanner 2 during office hour	office hour utilization of scanner 2
working time for scanner 1 during non-office hour	non-office hour utilization of scanner 1
total actual office hour	office hour utilization of scanners
total actual non-office hour	non-office hour utilization of scanners
date that outpatient calls	access times for outpatients
date that outpatient actual scanned	access times for outpatients
absolute actual clock time of current time	patients waiting time
absolute actual clock time of patient start scanning	patients waiting time
binary flag that if patient has no chair	the fraction of patients waiting outside the waiting room
binary flag that if patient is scanned outside office hour	the percentage cannot be scanned in office hour

TABLE 2: Data collected during the simulation

4.3 Confidence interval

We use batch mean method for steady state simulation. We assign the first 50 weeks (=50*7*24 = 8400 hours) to be the warm up length. The actual running time is 150 weeks. Why we choose the number od weeks this long is because the simulation is a long-term simulation. At first a warm up size of 12 weeks and 48 weeks for running batches is tried but cannot reach the precision of 10% we need. Because outpatient wait times are highly dependent on the random variable of probability 0.16 of not showing up and slot time will leave empty, the wait times are different from each other very significantly between batches. The check for exceed chair limitation using binary check also requires longer weeks.
After deleting this period we divide the reminder run into k=30 batches of approximately 5 weeks (= 35 days = 840 hours) . We decided on this number based the Rule of thumb that b=4D.
For all the measurements, let M_i be the mean metric value during batch i. Then the sample mean $\overline{M}$ equals

$$\overline{M} = \frac{\sum_{i=1}^{k} M_i}{k} \tag{11}$$

and the standard diviation is

$$\sqrt{\frac{1}{k-1}\sum_{i=1}^{k}(M_i - \overline{M})} \tag{12}$$

Because of student T distribution, with degree of freedom to be k-1=29, we have the critical $t_{\frac{\alpha}{2},k-1}$ value equals 2.04523. Then the half width of the confidence interval is

$$w = t_{\frac{\alpha}{2},k-1} \cdot \frac{\sqrt{\frac{1}{k-1}\sum_{i=1}^{k}(M_i - \overline{M})}}{\sqrt{k}} \tag{13}$$

Then we get the confidence interval $CI = [\overline{M} - w, \overline{M} + w]$ for every metric.

5 Results

The relative precision is depend on the performance measurement. We calculate the precision by

$$p = \frac{CI\ half\ width}{E[mean]} = \frac{w}{\overline{\overline{M}}} \tag{14}$$

5.1 Verification simulation M/M/2

According to the setting in section 3.3, with $\lambda = 1$ and enable the scanner 2 working for the whole day regardless of office hours. The mean waiting hours of (only emergency) patients is 0.00354 with confidence interval [0.00317,0.00391]. Since the final calculated theoretical result of M/M/2 waiting time is 0.00373, it fits in the confidence interval of the simulated waiting time value. So the correctness of our model is proved.

5.2 CT scanner utilization

The utilization is the fraction of time each scanner scanning, distinguished between office hours and outside office hours. A scanner cannot be utilized less than 0. The difference between the utilization of scanner 1 and 2 in the same working time is because in actual coding we will check the availability of scanner 2 first. This priority of assigning patients to scanner 2 will maximize the total utility by making more use of office hours. The precisions are all less than 0.1, under 10% relative precision.

In section 4.1, we calculated that the theoretical utilization of office hour and non office hour should be 0.861 and

scanner	mean utilization	confidence interval	precision
1(working time)	0.675	[0.666,0.685]	0.0083
1(non-working time)	0.333	[0.329,0.338]	0.0138
2(only working time)	0.784	[0.773,0.794]	0.0138

TABLE 3: CT scanner utilization

0.332. The actual simulated utilization of non working hour is the same with calculated value, but the utilization in office hour varies a lot with $\rho = \frac{0.675+0.784}{2} = 0.7295 < 0.861$. This is because the actual simulation includes the probability of outpatients not showing up.

5.3 Access times for outpatients

The mean number of days between the request and the actual appointment is 1.458 days, with confidence interval [1.429,1.486], precision 1.91%. This time is record in natural days between the outpatient's call and scheduled slot.

Although we have 28 slots and $\lambda_O = 23$ calls per day, The capacity seems to be enough of the demand to let the access time less than about one day. But the access time we calculated is based on natural days and the Friday batching rule could let the patients waiting weekend days extra. And also the out patients have probability 0.16 of not showing up which leave the time slot empty, decreasing the actual valid number of slots.

5.4 Waiting times for emergency patients and for out patients

patient type	waiting time	confidence interval	precision
emergency patients	0.054	[0.053,0.056]	0.0288
outpatients	0.178	[0.169,0.187]	0.0481

TABLE 4: Waiting times for emergency patients and for out patients

5.5 The fraction of (CT) patients

The overflow fraction is 0.113 with confidence interval [0.106,0.120], which means 11.3% of the patients of all types are expected to wait without the chairs. Because the waiting room only has 3 chairs, and patients are waiting for about 0.3 hours, the chairs are not enough when multiple appointments overlap with inpatient arrivals.

5.6 Percentage of inpatient delay

The percentage of those that cannot be scanned before 16:00 on the same weekday for inpatients with requests during office hours is 0.0022 with confidence interval [0.0011,0.0034], which means only 0.22% of inpatients whose requests arrive during office hours fail to get scanned before 16:00 on the same weekday.

The relative precision for this metric is 49.5%, much higher than the 10% precision we need. But The failure rate is actually low with a half width = 0.0011 which is a very small number. So we can see that there is measurement for an event that rarely happens. We may could improve the performance of this precision by making the simulation runs term long enough. This should be a positive result although it does not reach the 10% precision, it is high only because the precision is calculated relatively. While the relative precision is affected by the rarity of these events, the narrow absolute width of the confidence interval confirms that the probability of failure is always near zero.

Combining with the result in 5.4 we can see that although there is 11.29% of patients exceed out of the waiting room, the percentage of not finishing current work at a certain time point(end of office hour 16:00) is very low to be 0.22%. It shows that the problem is not the scanning work efficiency but is the number of chairs too few.

6 Conclusion

From preforming an analysis capacity check we found out that the capacity of the scanners is sufficient during both working hours and outside the working hours. We can also see, that it would not be sufficient to have only one CT scanner during the working hours.
During the working hours, one of the scanners is used 78.4% of the time on average and the other one for 67.5%. Outside of the working hours the scanner that remains active gets used 33.3% of the time.
Currently, after an outpatient calls for an appointment, the average time between the appointment and the call is 1.458 days. We can say that this is relatively fast, since we also need to count for the people who called on Friday and therefore have to wait for after the weekend.
The average waiting time for an emergency patient is 3.24 minutes and for an outpatient it is 10.68 minutes.
11.29% of all patients cannot wait inside of the waiting room. Therefore it might be beneficial to add another chair.
0.22% of inpatients whose requests arrive during office hours fail to get scanned before 16:00 on the same weekday.

7 Appendix

7.1 AI statement

No AI usage.